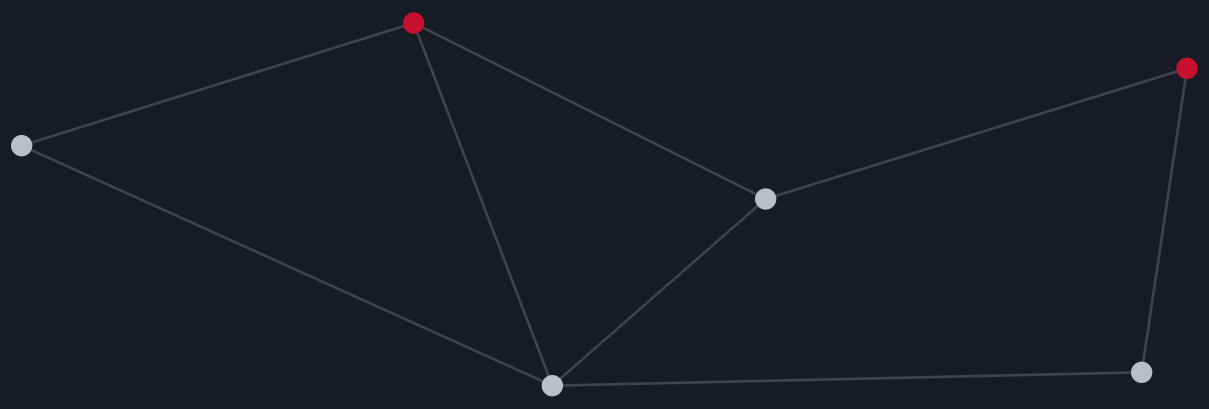
MISSO X Cougar CS X FITP

OCTOBER 23-24, 2026

CASE QUEST

PARTICIPANT PACKET

A business and technology case competition.
Your guide to the challenge, team experience, submission and judging.



SPONSORED BY

ConocoPhillips

YOUR EVENT GUIDE

Keep this packet alongside with you , judging rubric and dataset files.

03 THE CHALLENGE

Business decisions, technical design and your team

04 FRIDAY SCHEDULE

Case launch and team preparation

05 SATURDAY SCHEDULE

Final analysis, presentations and awards

06 WORKSHOPS

Interactive workshops and activities designed to develop practical skills throughout the event.

07 JUDGING/DELIVERABLES

What your team needs to deliver

The Challenge

Background: Lone Mesa Energy is a fictional oil producer operating six sites and eighteen pieces of equipment. Equipment failures, maintenance delays, spare-parts shortages, and safety concerns are disrupting production.

Your Challenge: Act as consultants to identify an operational problem and propose **one practical business or technology solution** that improves efficiency, reduces downtime, or strengthens workplace safety.

Use the provided **Assets, Events, Production, Readings, Solutions, and HSE datasets** to support your recommendation.

Guiding Questions

1. What problems should the company prioritize?
Identify the key major operational challenges using relevant data. Consider equipment downtime, production losses, costs, or safety concerns. Which is the highest priority?

2. What does the data reveal?
Analyze relevant datasets to identify trends, patterns, or inconsistencies. Support your findings with calculations or visualizations.

3. What solution would you recommend?
Propose an improvement such as equipment monitoring, maintenance scheduling, spare-parts management, or automation. Explain its benefits, estimated costs, and safety considerations.

4. How would your solution work?
Create a **functional prototype or interactive dashboard** using the provided data. Demonstrate how employees would use it to support operational decisions. Excel, Power BI, Tableau, Python, or web applications are acceptable.

5. How would the company implement the solution?
Outline the main implementation steps, resources, training needs, and measures of success or KPI's.

FRIDAY / October 23

Room: MH 160

AGENDA - All times are Central Time

TIME	ACTIVITY
10:00 - 10:30 AM	Check-in for CaseQuest (Mandatory)
10:30 - 11:00 AM	Welcome and Competition Format
11:00 – 11:30 pm	Case release and Group Matching
11:30 - 12:30 PM	Meal break
2:30 - 3:00 PM	Workshop:
4:00 - 4:30 PM	The Vibe Workshop
5:00 - 5:30 PM	FAQ
	GOOD LUCK!!

IMPORTANT: Discord will be our primary communication platform! All judging room assignments and important event updates will be shared there. Please make sure to check Discord regularly.

SATURDAY / October 24

Room: MH 160

AGENDA - All times are Central Time

TIME	ACTIVITY
1:00 pm	Check-in begins
1:00 - 2:00 PM	Meal break
2:30 - 3:00 PM	Industry Professional Pitch Workshop (At least 1 member per team required)
4:45 PM	Proposed submission deadline
4:45 - 5:00 PM	Check in for Judging (MH160)
5:00 - 6:30 PM	First Round Judging
6:30- 7:30 PM	Second Round Judging
End of Event	Proposed awards and closing

IMPORTANT: Discord will be our primary communication platform! All judging room assignments and important event updates will be shared there. Please make sure to check Discord regularly.

WORKSHOPS – working process will be done by Cougar CS and FITP

The Vibe Workshop

Room and Time: MH 160 | October 23 | 4:00 – 4:30 pm

Learn how to turn your ideas into interactive prototypes using AI-powered tools! Discover how to bring your concepts to life through hands-on prototyping and vibe coding, even with little to no coding experience.

The Pitch Workshop

Room and Time: MH 160 | October 24 | 2:30 – 3:00 pm

Learn how to confidently pitch your ideas and turn them into compelling presentations! Join us for an interactive workshop featuring **Kevin S., Product Development Analyst at NRG**, who will share industry insights and practical tips on how to effectively communicate and sell your ideas.

The TBA Workshop

Time:

HOW JUDGING WORKS

CRITERION	POINTS
Business problem and prioritization	15
Data analysis and evidence	20
Technology design and integration or dashboard	20
HSE, security and responsible use	15
Implementation and adoption	10
Recommendation, costs & resources	10
Presentation & clarity	10
TOTAL	100
Bonus: Data Storytelling and Credibility	5

Deliverables:

All teams are required to prepare and present a slide deck of no more than 10 slides, highlighting their key findings, data analysis, and proposed solutions. Each team will have a maximum of 7 minutes to present. Every team member must actively participate and be prepared to answer questions about their data, methodology, and recommendations.

Teams may also choose to develop a technical prototype or dashboard to support their proposed solution. Any technical prototype must be submitted alongside the final slide deck.